

# Wavelength Division Multiplexing (WDM)

Salwa El-Sabban

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## WDM

- Wavelength division multiplexing (WDM) is a technology used to combine or retrieve two or more optical signals of different optical center wavelengths in a fiber.
- It allows fiber capacity to be expanded in the frequency domain from one channel to more than 100 channels.
- This is accomplished by first converting standard, non-WDM optical signals to signals with unique WDM wavelengths that correspond to the available channel center wavelengths in the WDM multiplexer and demultiplexer.
- WDM channels are defined and labeled by their center wavelength or frequency and channel spacing.

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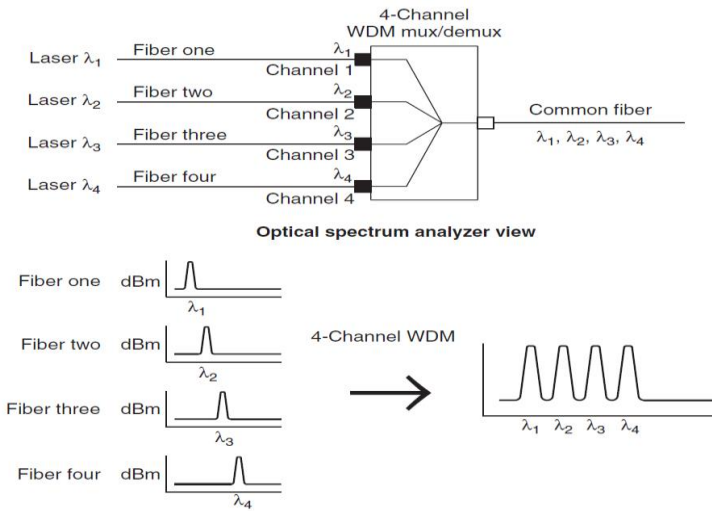
WDM

- The different WDM signal wavelengths are combined into one fiber by the WDM multiplexer.
- In the fiber, the individual signals propagate with little interaction assuming low signal power. For high powers, inter-channel interaction can occur.
- Once the signals reach the fiber link end, the WDM demultiplexer separates the signals by their wavelengths, back to individual fibers that are connected to their respective equipment receivers.
- The process of combining and separating the different wavelengths in the fiber is the basis of WDM technology.

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WDM

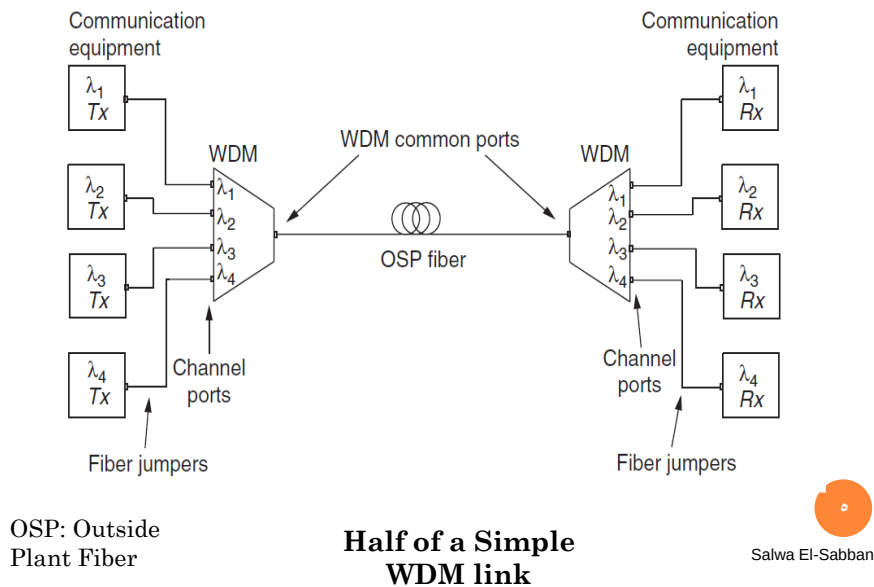


WDM functional block diagram

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## WDM



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## WDM-Filters

Four competing technologies are available for WDM filters: fiber Bragg grating (FBG), array waveguide grating (AWG), thin-film filter (TFF), and diffraction grating filter (DF):

- **Fiber Bragg gratings** have good filter shape but need to be used with circulators, which increases the WDM cost and size.
  - **Thin-film filters** have excellent filter response and low cost. They are typically used for WDMs with less than 40 channels.
  - **Array waveguide gratings** are suitable for larger channel counts and have good filter response.
  - **Diffraction gratings** are simple and suitable for WDM with large channel counts.
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WDM-Filters

Fiber Bragg gratings (FBG)

-A short length of fiber with a periodic change in its refractive index (known as a grating) works as a highly reflective mirror to a narrow band of wavelengths and passes all other wavelengths. The distance between two adjacent maximum values of the refractive index is referred to as the grating period.

-For a constructive interference between reflected waves from different grating period sections, the following condition must be achieved:

path difference= $m2\pi$

For  $m=1$

$2\beta\Lambda = 2\pi$

$\lambda_B=2n_{eff} \Lambda$

$\beta$ =propagation constant

Bragg wavelength

central reflection wavelength

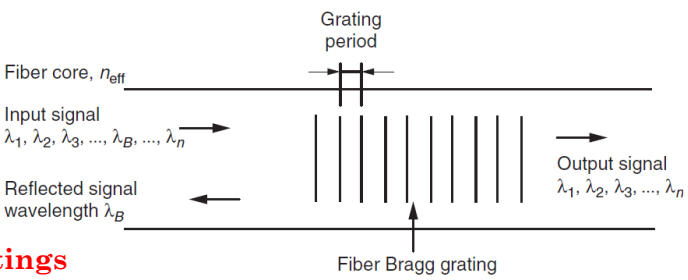
$n_{eff}$  = effective refractive index of the fiber core

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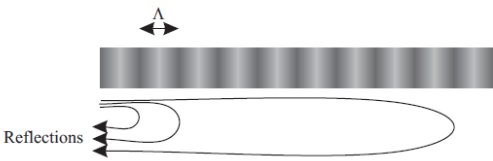
WDM-Filters

Bragg gratings



Bragg gratings

Principle of operation of Bragg gratings



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## WDM

### Single channel OADM using FBG and circulators

-A simple optical add/drop multiplexer OADM is constructed with one fiber Bragg grating and two circulators.

-The input signals to multiple channels  $\lambda_1$  to  $\lambda_n$  and channel  $\lambda_{B1}$  pass through the circulator and into the fiber Bragg grating. The grating reflects the  $\lambda_{B1}$  channel and passes all the others. The  $\lambda_{B1}$  channel reenters the circulator and exits at the drop port. Similarly, an added channel  $\lambda_{B2}$  is coupled to the add port and enters the circulator where it is directed into the fiber Bragg grating. The grating reflects it back into the circulator and to the output port.

-A similar kind of signal flow occurs for a **three-channel WDM Mux** and for a **three-channel WDM demux**.



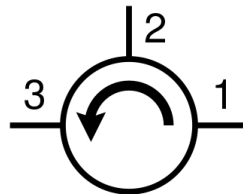
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## WDM-Filters

### Circulators

An **optical circulator** is a three- or four-port **optical device** designed such that **light** entering any port exits from the next. This means that if light enters port 1 it is emitted from port 2, but if some of the emitted light is reflected back to the circulator, it does not come out of port 1 but instead exits from port 3.

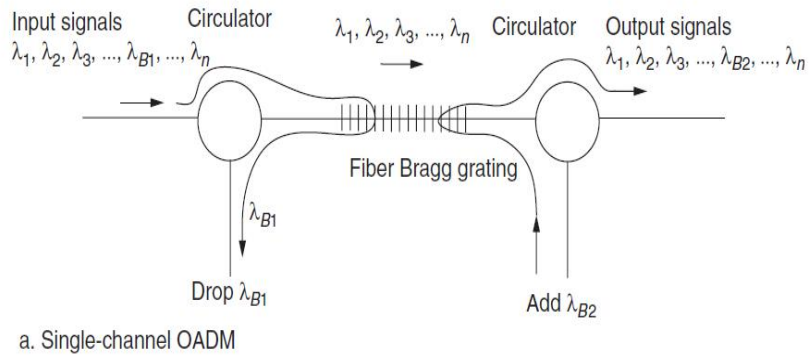


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## WDM-Filters

### Bragg gratings

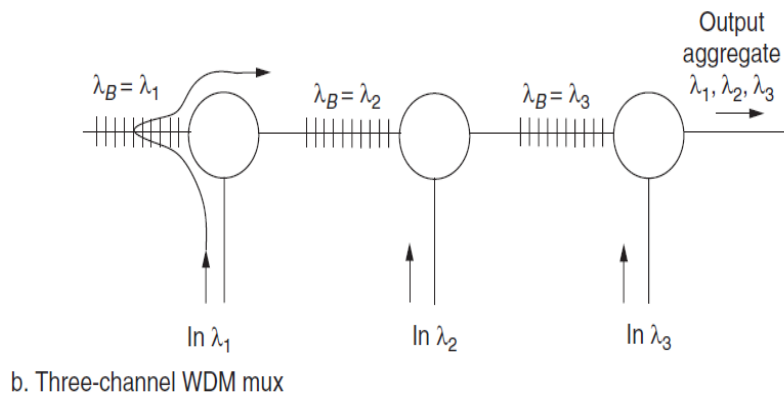


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## WDM-Filters

### Bragg gratings

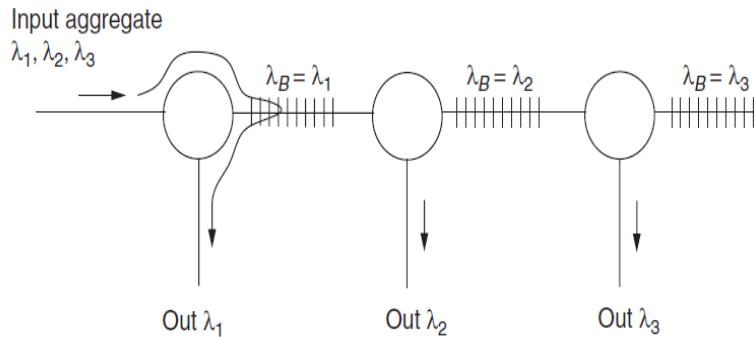


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## WDM-Filters

### Bragg gratings



c. Three-channel WDM demux



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## WDM-Filters (Bragg gratings)

1. FBGs have low insertion loss, typically less than 1 dB per filter.
2. FBG filters in WDMs are connected in series, which increases insertion loss as channel count increases.
3. FBG WDMs are available with channel spacing greater than 50 GHz.
4. FBGs are suitable for WDMs with less than 40 channels and greater than 50 GHz channel spacing. This is primarily due to accumulated loss caused by serial filters.



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## WDM-Filters

### Diffraction gratings (DF)

-A diffraction grating filter (DF) has a periodic structure that causes the incident light to be reflected or transmitted. For a reflective grating, periodic reflective ridges are formed on the surface of the filter. The grating reflects incident light into different angles corresponding to its component wavelengths.

-For a transmission grating, incident light passes through the grating and is diffracted at different angles corresponding to its different wavelength components.

-The channel fibers are properly positioned in the unit to receive the spatially separated wavelengths.

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## WDM-Filters

### Diffraction gratings (DF)

Each wavelength component is reflected at a direction where the reflection from each period of the grating add in phase (constructive interference) for this wavelength, i.e. the direction in which the path length difference for the reflections from neighboring periods of the grating is an integer number of wave lengths.

$$\begin{aligned}
 & k(\text{path length1}) - k(\text{path length2}) = m2\pi \\
 & m: \text{order of grating} \quad (\text{path length1}) - (\text{path length2}) = m\lambda \\
 & DC - AB = m \cdot \lambda \Rightarrow d \sin \Psi_1 - d \sin \Psi_0 = m \cdot \lambda
 \end{aligned}$$

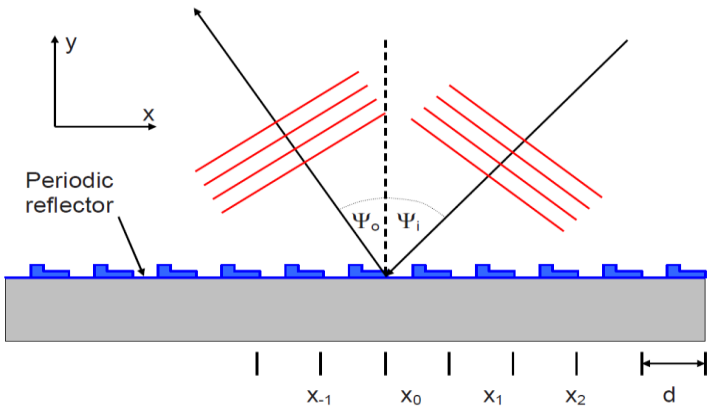
where  $\Psi_1$  and  $\Psi_0$  are the incident and diffracted angles,  $d$  is the grating period,  $\lambda$  is the wavelength of the incident monochromatic light, and  $m$  is any integer.

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WDM-Filters

Diffraction gratings (DF)



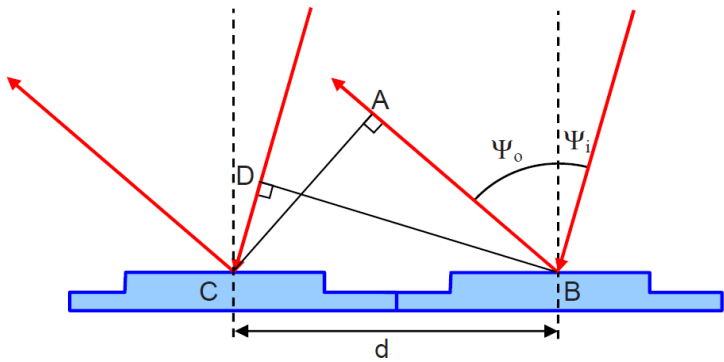
Schematic drawing of a periodic reflection grating.

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Diffraction gratings (DF)



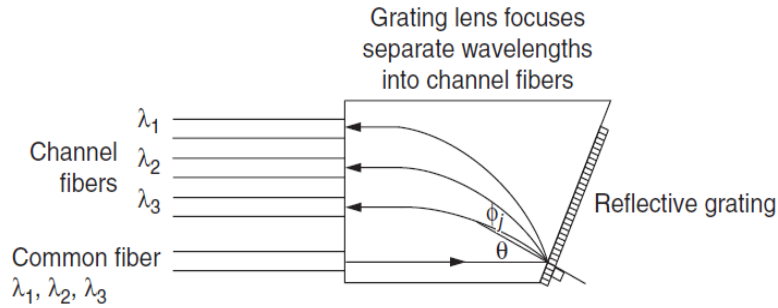
Schematic drawing of a periodic reflection grating.

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## WDM-Filters

### Diffraction gratings (DF)



a. Diffraction grating filter (DF) reflection demux

Reflective diffraction grating simple WDM

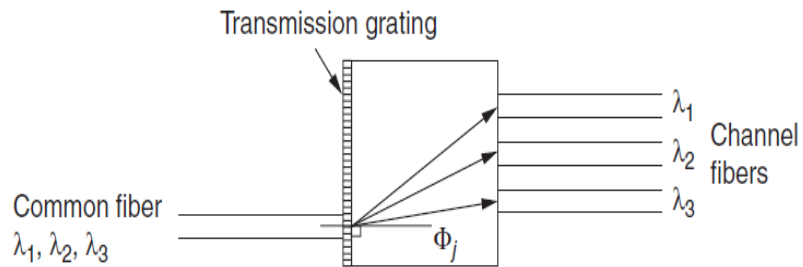
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### Diffraction gratings (DF)



b. Diffraction grating filter (DF) transmission demux

For this type of transmission diffraction grating

$$d \sin \Psi_0 = m \cdot \lambda$$

Transmission diffraction grating simple WDM

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## WDM-Filters

### Diffraction gratings (DF)

1. DFs have low insertion loss at high channel counts (~4 dB for 56 channels at 100 GHz)
2. DF WDMs are available with channel spacing greater than or equal to 50 GHz.
3. DFs are suitable for WDMs with large channel counts.



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## WDM-Filters

### Array waveguide gratings(AWG)

Array waveguide gratings(AWG) consists of  $M_{in}$  input and  $M_{out}$  output slab waveguides designated as regions 1 and 6 respectively. The slab waveguides interface to two identical focusing planar star couplers located in regions 2 and 5. An array of  $N$  uncoupled waveguides which have a propagation constant  $\beta$  connect the star couplers together. In the grating array region, the path length of each waveguide differs by a very precise amount  $\Delta L$  from the lengths in adjacent arms.

This device can function as a multiplexer, a demultiplexer, or a wavelength router (cross-connect switch).

For a pure multiplexer, we can take  $M_{out} = 1$ . For a demultiplexer,  $M_{in} = 1$ . In the case of a network routing application,  $M_{in} = M_{out}$ .

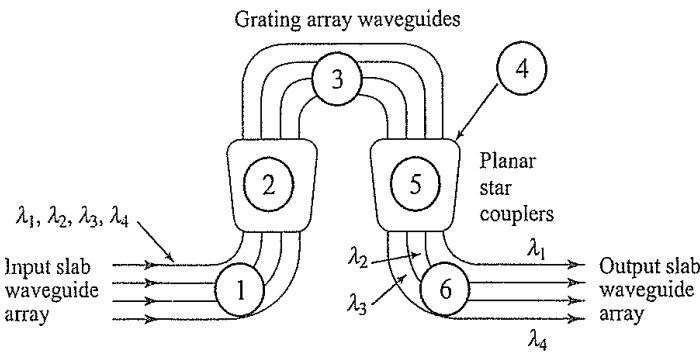


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Array waveguide gratings(AWG)



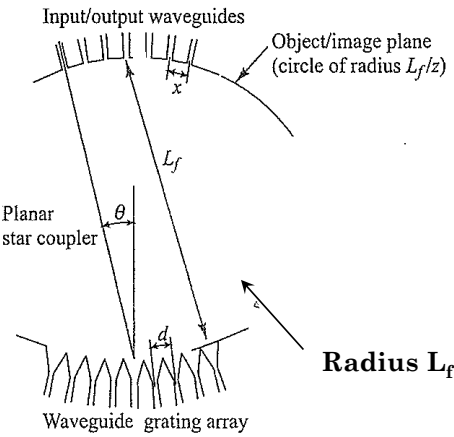
Top view of a typical arrayed waveguide grating and designation of its various operating regions

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WDM-Filters

Array waveguide gratings(AWG)

Geometry of the star coupler used in the arrayed waveguide grating WDM device



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## WDM-Filters

### Array waveguide gratings(AWG)

-The star coupler acts as a lens of focal length  $L_f$  so that the object plane is located at a distance  $L_f$  from the transmitter arrayed waveguides, and the image plane is located at a distance  $L_f$  from the receiver arrayed waveguides. The end of the arrayed waveguides are located on a part of a circle of radius  $L_f$ . Both the input and output waveguides are positioned on the focal lines, which are circles of radius  $L_f/2$ .

-In the previous figure, the parameter  $x$  is the center-to-center spacing between the input waveguides and the output waveguides at the star coupler interfaces,  $d$  is the spacing between the grating array waveguides, and  $\theta$  is the diffraction angle in the input or output slab waveguide.

-The refractive indices of the star coupler and the grating array waveguides are  $n_s$  and  $n_c$  respectively.

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## WDM-Filters

### Array waveguide gratings(AWG)

From a top-level viewpoint, the AWG functions as follows:

- Starting from the left, the input slab waveguides in region 1 are connected to a planar star coupler (region 2) which acts as a lens.
- The lens distributes the entering optical power among the different waveguides in the grating array in region 3.
- Adjacent waveguides of the grating array in region 3 differ in path length by a precise length  $\Delta L$ . The path length differences  $\Delta L$  can be chosen such that all input wavelengths emerge at point 4 with different phase delays  $\Delta\Phi \approx 2\pi n_c \Delta L / \lambda$ .

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## WDM-Filters

### Array waveguide gratings(AWG)

From a top-level viewpoint, the AWG functions as follows (continue):

- The second lens in region 5 refocuses the light from all the grating array waveguides onto the output slab waveguide array in region 6.
- As the phase-front will be different for each wavelength, the propagation direction will be different. (In another way, the constructive interference between light from different arrayed waveguides is achieved at a different output position depending on the wavelength)
- Thus, each wavelength is focused into a different output waveguide in region 6.

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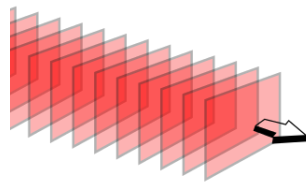
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## WDM-Filters

### Wavefront (Phase-front)

The wavefront of a time-varying field is the set (locus) of all points where the wave has the same phase.

The direction of propagation is perpendicular to the phase-front.



Plane wave example (wavefront perpendicular on propagation direction)

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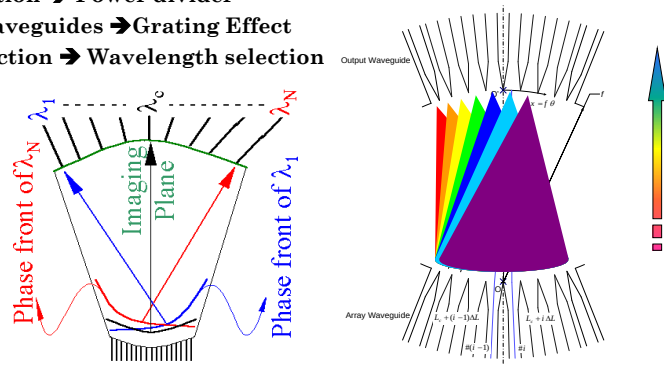
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## WDM-Filters

### Array waveguide gratings(AWG)

#### Basic Idea of Operation

- First section → Power divider
- Array Waveguides → Grating Effect
- Third Section → Wavelength selection



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## WDM-Filters

### Array waveguide gratings(AWG)

- From the phase-matching condition, the light emitted from the output channel waveguides must satisfy the grating equation: (constructive interference)

$$\Delta L 2\pi n_c / \lambda + d \sin \theta 2\pi n_s / \lambda = m 2\pi$$

Difference in phase  
from the waveguide  
section

Difference in phase  
from the star coupler  
section

$$\Delta L n_c + d \sin \theta n_s = m \lambda \quad (1)$$

where the integer  $m$  is the diffraction order of the grating

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## WDM-Filters

### Array waveguide gratings(AWG)

- Focusing is achieved by making the path-length difference  $\Delta L$  between adjacent array waveguides, measured inside the array, to be an integer multiple of the central design wavelength of the demultiplexer

$$\Delta L = m \frac{\lambda_c}{n_c}$$

where  $\lambda_c$  is the central wavelength in vacuum; that is, it is defined as the pass wavelength for the path from the center input waveguide to the center output waveguide.

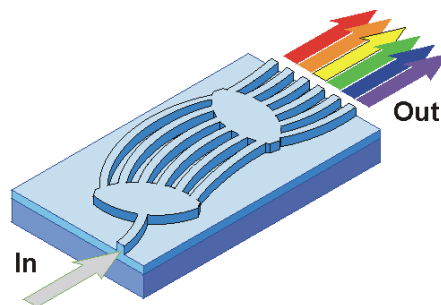
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Array waveguide gratings(AWG) (sometimes called Phased arrayed waveguide grating PHASAR)



Arrayed WG Grating as a **demultiplexer**

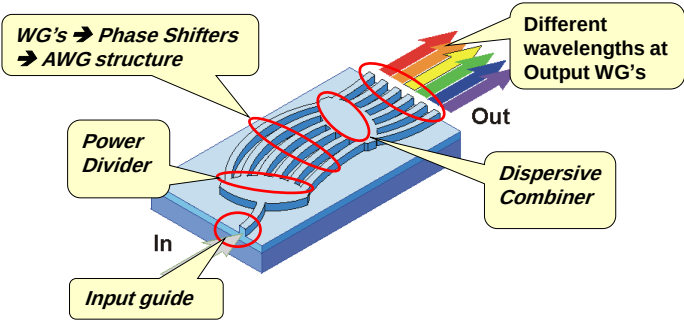
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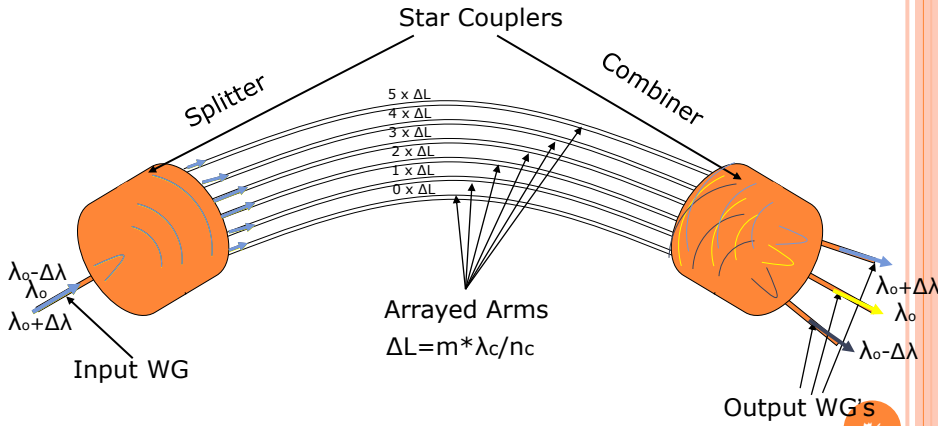
Array waveguide gratings(AWG)



AWG: STRUCTURE

WDM-Filters

Array waveguide gratings(AWG)



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### Array waveguide gratings(AWG)

To determine the channel spacing, we need to find the angular dispersion. This is defined as the incremental lateral displacement of the focal spot along the image plane per unit frequency change and is found by differentiating Eq. (1) with respect to frequency.

$$\text{(for small } \theta \rightarrow \frac{d\theta}{d\nu} = -\frac{m\lambda^2}{n_s c d} \frac{n_g}{n_c}$$

where the group index of the grating array waveguide is defined as

$$n_g = n_c - \lambda \frac{dn_c}{d\lambda}$$

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### Array waveguide gratings(AWG)

For small  $\Delta\theta$  :  $\tan \Delta\theta = \sin \Delta\theta = \Delta\theta = x/L_f$

→ in terms of **frequency**, the channel spacing  $\Delta\nu$  is

$$\Delta\nu = \frac{x}{L_f} \left( \frac{d\theta}{d\nu} \right)^{-1} = \frac{x \cdot n_s c d}{L_f m \lambda^2} \frac{n_c}{n_g}$$

or, in terms of **wavelength**,

$$\Delta\lambda = \frac{x}{L_f} \frac{n_s d}{m} \frac{n_c}{n_g} = \frac{x}{L_f} \frac{\lambda_0 d}{\Delta L} \frac{n_s}{n_g}$$

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Array waveguide gratings(AWG)

Last two Equations thus define the pass frequencies or wavelengths for which the multiplexer operates, given that it is designed for central wavelength  $\lambda_c$  We also note that by making  $\Delta L$  large, the device can multiplex and demultiplex optical signals with very small wavelength spacings.

Due to the periodicity of equation 1 ( $m=1, 2, 3, 4, \dots$ )  
→ after every change of  $2\pi$  in the phase shift between adjacent waveguides, the field will again be imaged at the same spot and higher order wavelengths could be transmitted in the same output guide.

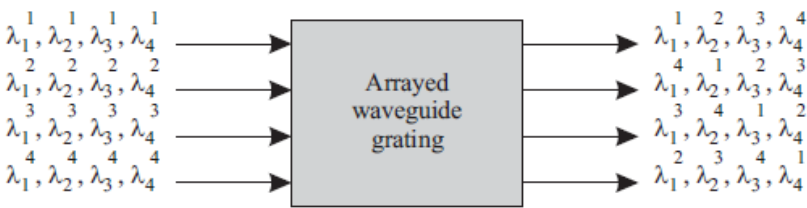
The difference between two successive wavelengths in the same output guide is called free spectral range (FSR)

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WDM-Filters

Array waveguide gratings(AWG)

The AWG can also be used as a static wavelength optical cross-connect (OXC).



4x 4 static wavelength cross connect using four wavelengths with one wavelength routed from each of the inputs to each of the outputs.

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## WDM-Filters

### Array waveguide gratings(AWG)

Main characteristics of AWG technology are as follows:

1. AWGs have a constant insertion loss of 4 to 5 dB relatively independent of the number of channels.
2. AWG WDMs are available with channel spacing of 12.5 to 200 GHz.
3. AWGs are suitable for WDMs with large channel counts (>40 channels).

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## WDM-Filters

### Thin-film filters (TFF)

- A thin-film resonant cavity filter (TFF) is a Fabry-Perot interferometer, or etalon, where the mirrors surrounding the cavity are realized by using multiple reflective dielectric thin-film layers.
- Multiple reflective dielectric thin-film layers structure contains multiple layers of very thin high- and low-refractive index materials
- Using multiple reflective dielectric thin-film layers works as a highly reflective mirror to a narrow band of wavelengths and passes all other wavelengths (as in Bragg grating).
- The thickness of the layers determines which wavelengths will be reflected and which will be transmitted through the filter.

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## WDM-Filters

### Thin-film filters (TFF)

- A **thin-film resonant cavity filter** acts as a bandpass filter that passes a narrow band of optical spectrum and reflects the rest.
- The wavelength that is passed through is determined by the cavity length.
- A thin-film resonant **multicavity** (coupled cavities) filter consists of two or more cavities separated by reflective dielectric thin-film layers. As more cavities are added, the top of the passband becomes flatter with steep rollofs.

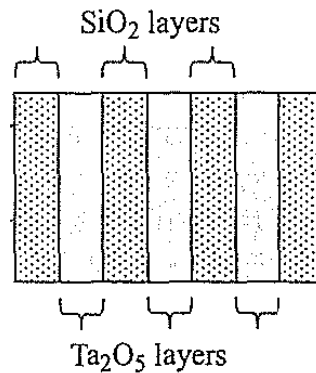
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## WDM-Filters

### Thin-film filters (TFF)



An example of the materials used in multiple reflective dielectric thin-film layers

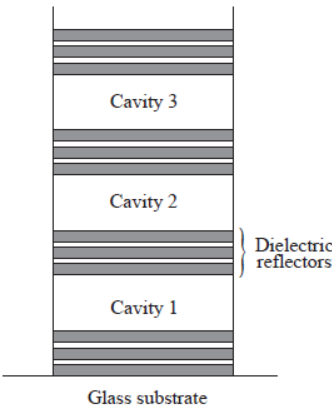
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Thin-film filters (TFF)

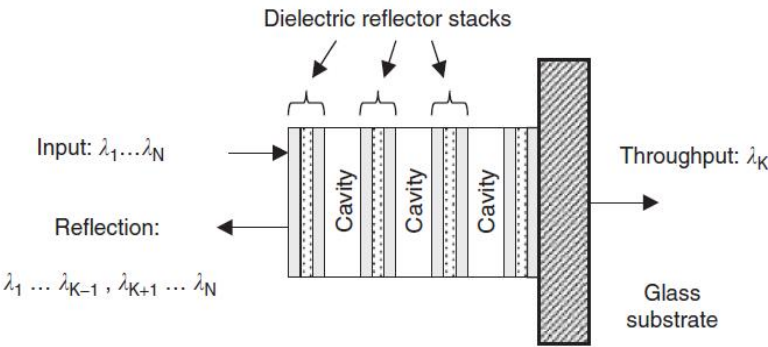


A three-cavity thin-film resonant dielectric thin-film filter.

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Thin-film filters (TFF)

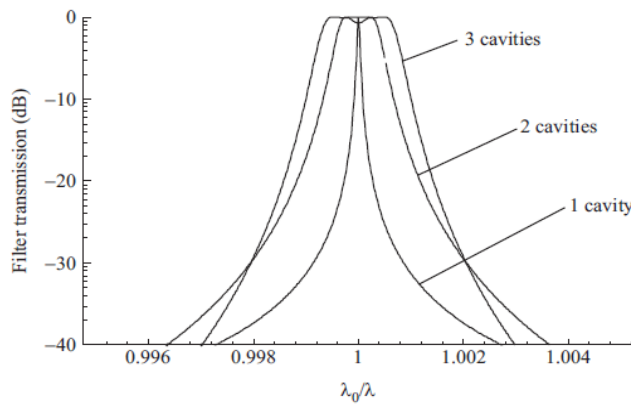


A multilayer optical thin-film filter consists of a stack of several dielectric thin films separated by cavities

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## WDM-Filters

### Thin-film filters (TFF)



Transfer functions of single-cavity, two-cavity, and three-cavity dielectric thin-film filters. Note how the use of multiple cavities leads to a flatter passband and a sharper transition from the passband to the stop band.

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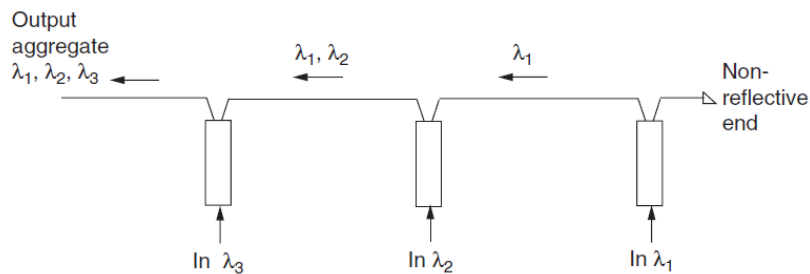
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## WDM-Filters

### Thin-film filters (TFF)

Three-channel TFF WDM mux



$\lambda_1$  is transmitted through the 1<sup>st</sup> TFF and then reflected from the 2<sup>nd</sup> TFF while the  $\lambda_2$  is transmitted through the 2<sup>nd</sup> TFF.  $\lambda_1, \lambda_2$  are reflected from the 3<sup>rd</sup> TFF while  $\lambda_3$  is transmitted through 3<sup>rd</sup> TFF.

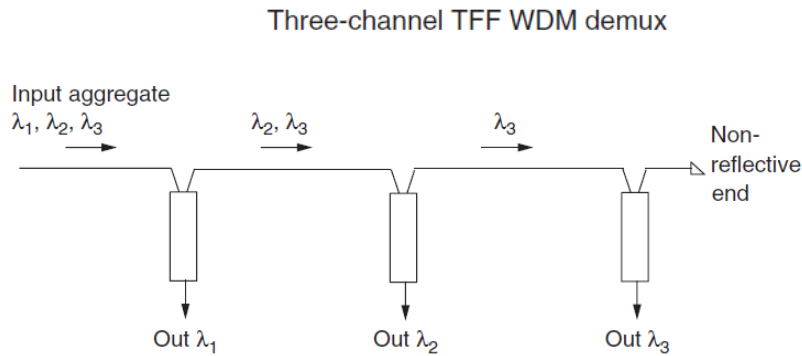
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## WDM-Filters

### Thin-film filters (TFF)



$\lambda_1$  is transmitted through the 1<sup>st</sup> TFF and  $\lambda_2, \lambda_3$  are reflected.  $\lambda_2$  is transmitted through the 2<sup>nd</sup> TFF while  $\lambda_3$  is reflected.  $\lambda_3$  is transmitted through 3<sup>rd</sup> TFF.

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## WDM-Filters (Thin-film filters)

Main characteristics of TFF technology are as follows:

1. TFFs have low insertion loss, typically less than 1 dB per filter.
2. TFFs in WDMs are connected in series, which increases insertion loss as channel count increases.
3. TFF WDMs are available with channel spacing greater than 50 GHz.
4. TFFs can be made tunable by changing the incident light angle of incidence.
5. TFFs are suitable for WDMs with less than 40 channels and greater than or equal to 50 GHz channel spacing.

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## WDM-Filters

### Important Note:

-For a transmission multilayer structure, the width of each layer =  $\lambda_{\text{medium}}/2$

where  $\lambda_{\text{medium}}$  is the transmitted wavelength

-For a reflection multilayer structure, the width of each layer =  $\lambda_{\text{medium}}/4$

where  $\lambda_{\text{medium}}$  is the reflected wavelength

where  $\lambda_{\text{medium}} = \lambda_o/n$



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